

Internet of Things and its Applications

Chapter 2

Things and Connections

Control Systems • OSI & TCP/IP Models • Transmission Modes
Wired & Wireless Media • IoT Process Flow

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1. Working of a Control System

EN A **control system** is an arrangement of mechanical or electronic components that manages the behaviour of another device or process through a control loop. In IoT, most control systems are computer-based. Different industries use control systems to regulate machinery, keep output within a desired range, and reduce manual intervention.

HI कंट्रोल सिस्टम यंत्रों और इलेक्ट्रॉनिक भागों का ऐसा समूह है जो एक कंट्रोल लूप के ज़रिए किसी दूसरी डिवाइस या प्रोसेस के व्यवहार को नियंत्रित करता है। IoT में अधिकांश कंट्रोल सिस्टम कंप्यूटर आधारित होते हैं। अलग-अलग उद्योगों में मशीनों को नियंत्रित करने, आउटपुट को तय सीमा में रखने और मैन्युअल हस्तक्षेप घटाने के लिए इनका उपयोग किया जाता है।

Common elements that form a control loop include **Programmable Logic Controllers (PLC)**, **Programmable Automation Controllers (PAC)**, **Remote Terminal Units (RTU)**, **Intelligent Electronic Devices (IED)**, and **Sensors**. A control loop typically consists of a sensor, a controller, and a final control element working together to measure, compare, and correct a process variable.

Key features of a control system: Accuracy, Energy saving, Improved safety, Stability, Speed.

1.1 Open Loop Control System

An open loop system does not use any feedback — it simply produces an output based on the input without checking whether the actual result matches the desired one. This makes it simple and cheap, but inaccurate if disturbances occur.

ओपन लूप सिस्टम में कोई फीडबैक नहीं होता — यह सिर्फ इनपुट के आधार पर आउटपुट देता है और यह जांच नहीं करता कि परिणाम वांछित परिणाम से मेल खाता है या नहीं। इसलिए यह सरल और सस्ता है, लेकिन गड़बड़ी होने पर अशुद्ध हो सकता है।

Example: A simple room heater — it keeps heating as long as it is switched on, regardless of the actual room temperature, because there is no sensor feeding data back to it. Other examples: table fan, toaster.

Open Loop Control System (No Feedback)



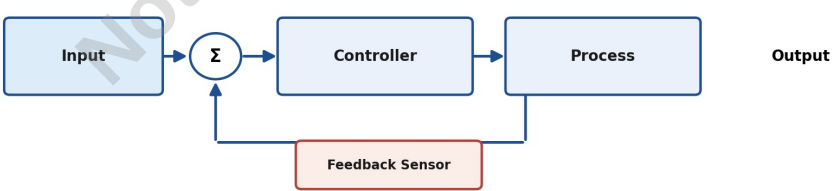
1.2 Closed Loop Control System

A closed loop (feedback) system continuously measures its output, compares it with the desired value, and adjusts itself until the difference (error) becomes zero. This makes it far more accurate than an open loop system, though more complex to build.

क्लोज्ड लूप (फीडबैक) सिस्टम लगातार अपने आउटपुट को मापता है, उसे वांछित मान से तुलना करता है, और तब तक समायोजन करता है जब तक अंतर (त्रुटि) शून्य न हो जाए। इससे यह ओपन लूप सिस्टम से कहीं अधिक सटीक बनता है, हालांकि इसका निर्माण अधिक जटिल होता है।

Example: A home air-conditioner — you set a target temperature; a sensor keeps checking the room temperature and sends feedback so the AC keeps cooling only until that target is reached.

Closed Loop Control System (With Feedback)



1.3 Open vs Closed Loop — Comparison

Open Loop Control System	Closed Loop Control System
No feedback element	Feedback element always present
No error detector	Error detector always present
Generally stable	Can become unstable if not tuned well
Simple to build	Complex to build
Low cost	Higher cost
Less accurate	More accurate
Less maintenance	More maintenance

1.4 Real-Time System

A real-time system performs specific tasks within strict time limits — the correctness of the output depends not just on the result but also on the time at which it is delivered. Typical examples are air-traffic control systems, industrial process-control systems, and autonomous-driving systems.

रीयल-टाइम सिस्टम विशिष्ट कार्यों को तय समय-सीमा के भीतर पूरा करता है — यहां परिणाम की सत्यता केवल आउटपुट पर ही नहीं बल्कि उसे कितने समय में दिया गया, इस पर भी निर्भर करती है। उदाहरण: हवाई यातायात नियंत्रण प्रणाली, प्रक्रिया नियंत्रण प्रणाली, स्वचालित (autonomous) ड्राइविंग सिस्टम।

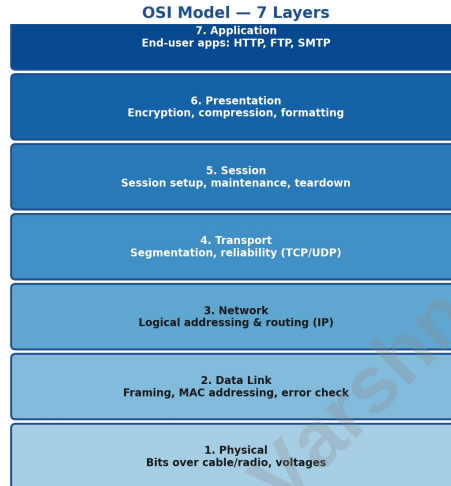
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2. Connectivity Models — OSI and TCP/IP

2.1 OSI Model

EN The **OSI (Open System Interconnection)** model was developed by **ISO** in **1984** as a conceptual reference framework describing how two systems communicate over a network. It is not implemented directly in real networks — it is used mainly as a reference to understand and design networking protocols. It divides the communication process into **7 independent layers**, each handling a specific job, and passes data from one layer to the next.

HI OSI मॉडल को ISO ने 1984 में एक संदर्भ (reference) मॉडल के रूप में विकसित किया, जो बताता है कि नेटवर्क में दो सिस्टम आपस में कैसे संवाद (communicate) करते हैं। इसे वास्तविक नेटवर्क में सीधे लागू नहीं किया जाता — इसका उपयोग नेटवर्किंग प्रोटोकॉल को समझने और डिज़ाइन करने के लिए संदर्भ के रूप में होता है। इसमें संचार प्रक्रिया को 7 अलग-अलग परतों (layers) में बांटा गया है।



#	Layer	Main Job
7	Application	Interface for end-user apps: HTTP, FTP, SMTP, NFS
6	Presentation	Data formatting, encryption/decryption, compression
5	Session	Establishes, manages and terminates sessions between devices
4	Transport	Breaks data into segments, ensures reliable delivery (TCP/UDP)
3	Network	Logical addressing (IP) and routing packets to destination
2	Data Link	Frames data, adds MAC address, basic error checking
1	Physical	Transmits raw bits as electrical/light/radio signals

Physical Layer — deals with the actual electrical/physical connection: voltages, cabling, data rate, and network topology; also decides whether the medium is wired or wireless.

Data Link Layer — organizes bits from the physical layer into **frames**, adds MAC addressing, and checks for transmission errors. Common protocols: HDLC, PPP.

Network Layer — responsible for logical (IP) addressing and choosing the best path (routing) for a packet from source to destination.

Transport Layer — splits messages into segments, numbers them for reassembly, and offers connection-oriented (TCP) or connectionless (UDP) delivery with flow/error control.

Session Layer — acts like a "dialog controller"; opens, maintains and closes a session between two communicating devices, and re-synchronizes after errors.

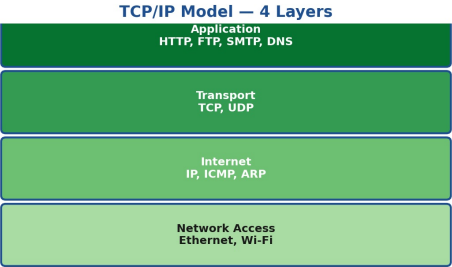
Presentation Layer — handles encryption for privacy and compression to reduce data size; also converts data into a format both systems understand.

Application Layer — closest to the end user; lets applications access network services such as file transfer, email, and database access.

2.2 TCP/IP Model

EN **TCP/IP (Transmission Control Protocol / Internet Protocol)** is the practical protocol suite that actually runs the internet. It was developed by the **US Department of Defense** between the 1970s and 1980s and provides **end-to-end** communication over interconnected networks. Unlike OSI's 7 layers, TCP/IP condenses functionality into **4 layers**.

HI TCP/IP वह व्यावहारिक प्रोटोकॉल समूह है जिस पर वास्तव में इंटरनेट चलता है। इसे 1970-80 के दशक में अमेरिकी रक्षा विभाग (DOD) ने विकसित किया और यह इंटरकनेक्टेड नेटवर्कों में एंड-टू-एंड संचार प्रदान करता है। OSI की 7 परतों की तुलना में TCP/IP केवल 4 परतों में काम करता है।



Layer	Function	Related OSI Layers
Application	User-facing services: web, email, file transfer (HTTP, FTP, SMTP, DNS)	Application, Presentation, Session
Transport	End-to-end delivery, reliability, flow control (TCP, UDP)	Transport
Internet	Logical addressing and routing of IP datagrams (IP, ICMP, ARP)	Network
Network Access	Delivers frames on the local network; maps IP to physical (MAC) address	Data Link, Physical

2.3 OSI vs TCP/IP

OSI Model	TCP/IP Model
Full form: Open System Interconnection	Full form: Transmission Control Protocol / Internet Protocol
Developed by ISO	Developed by ARPANET (US DOD)
7 layers	4 layers
Purely a reference/conceptual model	Practical model, used in real networks
Rarely used directly	Widely used — basis of the internet

3. Transmission Modes

EN The **transmission mode** (also called communication mode) describes the direction in which data flows between two devices. There are three types: **Simplex**, **Half-Duplex**, and **Full-Duplex**.

HI ट्रांसमिशन मोड (जिसे कम्युनिकेशन मोड भी कहते हैं) यह बताता है कि दो डिवाइसों के बीच डेटा किस दिशा में प्रवाहित होता है। इसके तीन प्रकार होते हैं: सिम्पलेक्स, हाफ-डुप्लेक्स और फुल-डुप्लेक्स।

Simplex — one direction only (e.g. Radio, TV)



Half-Duplex — both directions, one at a time (e.g. Walkie-talkie)



Full-Duplex — both directions simultaneously (e.g. Telephone)

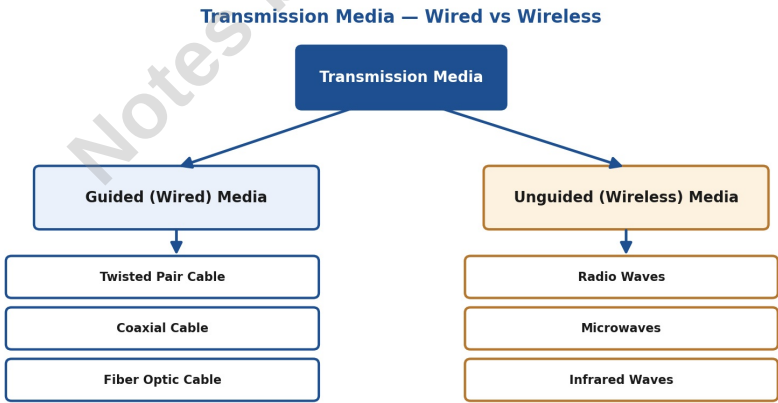


Mode	Description	Example
Simplex	Data flows in only one direction; sender cannot receive	Radio, TV broadcast, remote control
Half-Duplex	Data can flow both ways, but only one direction at a time	Walkie-talkie
Full-Duplex	Data flows both ways simultaneously — fastest mode	Telephone call

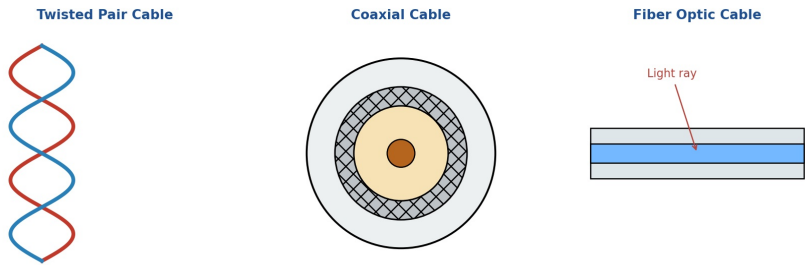
4. Wired & Wireless Transmission Media

EN Transmission media are classified into **Guided (Wired)** media, where a physical cable carries the signal, and **Unguided (Wireless)** media, where signals travel through the air as electromagnetic waves.

HI ट्रांसमिशन मीडिया को दो भागों में बांटा जाता है: गाइडेड (वायर्ड) मीडिया, जिसमें सिग्नल किसी भौतिक केबल से गुजरता है, और अनगाइडेड (वायरलेस) मीडिया, जिसमें सिग्नल हवा में विद्युत-चुंबकीय तरंगों (electromagnetic waves) के रूप में यात्रा करता है।



4.1 Wired (Guided) Media



Cable	Description	Typical Use
Twisted Pair	Two insulated copper wires twisted together to reduce signal interference (crosstalk)	Telephone lines, LAN (UTP/STP), can carry data over ~10 km at low-to-moderate speed
Coaxial Cable	A central copper core, insulation, a braided metal shield, and an outer jacket — carries stronger signals than twisted pair	Cable TV networks, CCTV
Fiber Optic	Sends data as pulses of light through a thin glass/plastic core; extremely fast and immune to electrical interference	Long-distance, high-speed backbone links, broadband internet

4.2 Wireless (Unguided) Media

Wireless media use **electromagnetic waves** to carry signals through space without any physical cable. This includes radio waves, microwaves, and infrared waves — each suited to a different range and use case.

वायरलेस मीडिया विद्युत-चुंबकीय तरंगों (electromagnetic waves) का उपयोग करके बिना किसी भौतिक केबल के अंतरिक्ष/हवा के माध्यम से सिग्नल भेजता है। इसमें रेडियो तरंगें, माइक्रोवेव और इन्फ्रारेड तरंगें शामिल हैं — हर एक की सीमा और उपयोग अलग-अलग होता है।

Wave Type	Frequency Range	Typical Use
Radio Waves	3 KHz – 300 GHz	Broadcast radio, long-range wireless communication; penetrates objects, travels long distances
Microwaves	300 MHz – 300 GHz	Long-distance point-to-point links, satellite communication; low cost
Infrared Waves	Above visible light frequency	Short-range use — TV remote, mobile data sharing; cannot pass through solid objects

4.3 Wired vs Wireless — Comparison

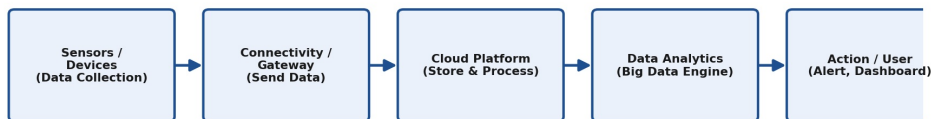
Wired Network	Wireless Network
Higher transmission speed	Comparatively slower transmission speed
Low propagation delay	Higher propagation delay
More secure and reliable	Comparatively less secure
Devices must be physically cabled	Quick to install, no cabling needed
Lower device cost, higher installation effort	Higher device cost, faster installation
Uses hubs, switches	Uses wireless routers, access points

5. Process Flow of an IoT Application

EN The Internet of Things connects everyday devices to the internet so they can sense their surroundings and exchange data automatically, without needing constant human involvement.

HI इंटरनेट ऑफ थिंग्स रोजमर्रा की डिवाइसों को इंटरनेट से जोड़ता है ताकि वे अपने आसपास के वातावरण को समझ सकें और बिना लगातार मानवीय हस्तक्षेप के आपस में डेटा का आदान-प्रदान कर सकें।

IoT Application — Process Flow



1. **Sense:** Sensors/devices collect data from the surrounding environment (temperature, motion, light, etc.).
2. **Connect:** This data is sent through a communication medium (Wi-Fi, cellular, LPWAN, etc.) via a gateway to the internet.
3. **Store/Process:** The data reaches a cloud platform where it is securely stored.
4. **Analyze:** A big-data analytics engine processes the data to detect patterns and support decision-making.
5. **Act:** The processed insight is delivered to the end user as an alert, email, dashboard display, or automated action.

पहले सेंसर/डिवाइस वातावरण से डेटा इकट्ठा करते हैं, फिर यह डेटा गेटवे और नेटवर्क के ज़रिए क्लाउड प्लेटफ़ॉर्म तक भेजा जाता है, जहाँ इसे सुरक्षित रूप से स्टोर किया जाता है। इसके बाद बिग डेटा एनालिटिक्स इंजन इसका विश्लेषण करता है, और अंत में परिणाम को ईमेल, डिस्प्ले या अलर्ट के माध्यम से एंड-यूजर तक पहुंचाया जाता है।

Chapter 2 — Practice Questions

Original practice questions for self-revision (not from any external source).

- Q1.** Which type of control system operates without comparing its output to a desired value?
a) Closed loop b) Open loop c) Feedback loop d) None of these
- Q2.** A system that continuously measures output and adjusts itself using an error signal is called:
a) Open loop system b) Closed loop system c) Static system d) None of these
- Q3.** Which of the following is an example of a closed loop control system?
a) Hand drier b) Table fan c) Air conditioner with thermostat d) Toaster
- Q4.** How many layers does the OSI model have?
a) 4 b) 5 c) 7 d) 9
- Q5.** Which OSI layer is responsible for logical (IP) addressing and routing?
a) Data Link b) Network c) Transport d) Session
- Q6.** The TCP/IP model has how many layers?
a) 3 b) 4 c) 5 d) 7
- Q7.** Which organization developed the OSI reference model?
a) IEEE b) ISO c) IETF d) W3C
- Q8.** Which transmission mode allows data flow in both directions at the same time?
a) Simplex b) Half-duplex c) Full-duplex d) None of these
- Q9.** A walkie-talkie is an example of which transmission mode?
a) Simplex b) Half-duplex c) Full-duplex d) None of these
- Q10.** Which of these is NOT a guided (wired) transmission medium?
a) Twisted pair cable b) Coaxial cable c) Fiber optic cable d) Microwave
- Q11.** The core of a fiber optic cable is generally made of:
a) Copper b) Aluminium c) Glass d) Rubber
- Q12.** Which cable is most commonly used for Cable TV and CCTV networks?
a) Twisted pair b) Coaxial cable c) Fiber optic d) None of these
- Q13.** Which wireless wave type is used in TV remote controls?
a) Radio waves b) Microwaves c) Infrared waves d) X-rays
- Q14.** In the IoT process flow, which stage comes immediately after data collection by sensors?
a) Analytics b) Sending data via connectivity/gateway c) User alert d) None of these
- Q15.** Which protocol pair operates at the Transport layer of TCP/IP?
a) HTTP, FTP b) TCP, UDP c) IP, ARP d) Ethernet, Wi-Fi